Forces: Particles in Motion

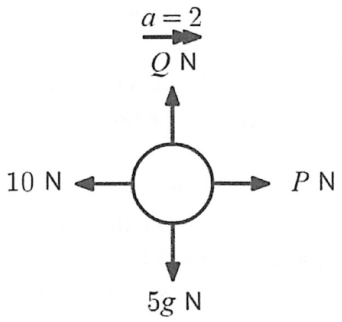
Questions 1–18

Question 1. A particle mass m kg is acted upon by a resultant force F N has acceleration a ms^{-2}. In each case, find the unknown variable:

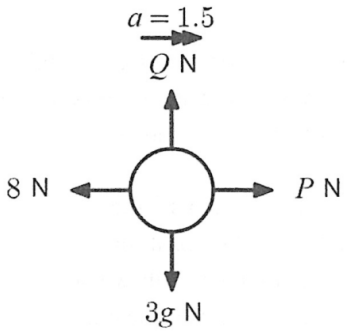
a) $m = 0.8$ kg, $a = 1.6$
b) $a = 0.8$, $F = 20$
c) $m = 5$ kg, $F = 15$
d) $m = 200$ g, $a = 0.8$

Question 2. The following questions use $g = 9.8$ **ms**$^{-2}$ A particle of mass m kg with forces acting on it as shown in the diagram below, is accelerating at a ms^{-2}. In each case, state the value of m and find the values of P and Q

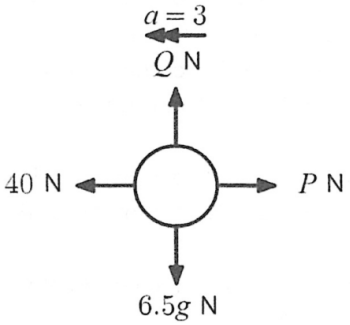
a)



b)

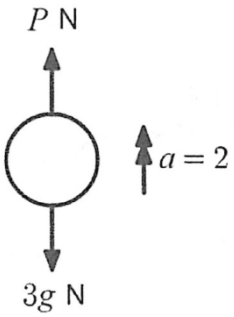


c)

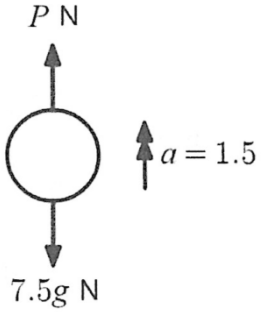


Question 3. The following questions use $g = 10$ **ms**$^{-2}$ A particle of mass m kg with forces acting on it as shown in the diagram below, is accelerating at a ms^{-2}. In each case, state the value of m and find the value of P

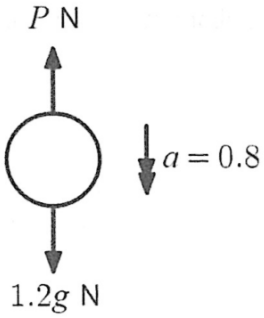
a)



b)



c)



Question 4. A particle of mass 2 kg is attached to the end of a light, vertical string. The particle is accelerating vertically at a ms^{-2}. In each case find the tension in the string when:

a) $a = 3$ ms^{-2} upwards.
b) $a = 1.5$ ms^{-2} downwards.
c) $a = -2.5$ ms^{-2} upwards.
d) moving with constant velocity.

Question 5. A particle of mass 0.5 kg is attached to the end of a light, vertical string. The particle is decelerating upwards at 0.2 ms^{-2}. Find the tension in the string.

Question 6. The following questions use $g = 9.8$ $\mathbf{ms}^{-2}$ A block of mass 8 kg is moving in a straight line with constant acceleration 0.1 ms^{-2}. The force driving it forward is 12 N. Find the total resistance acting against the motion of the block.

Question 7. A box is being pushed on a horizontal surface such that it is moving with constant velocity. The pushing force is 100 N. Find the total resistive force.

Question 8. A box is being pushed on a horizontal smooth surface by a 24 N force so that it is accelerating at 1.5 ms^{-2}.

a) Find the weight of the box.

b) State a modelling assumption used in your calculation.

Question 9. A lorry of mass 3 tonnes is driving along a straight road. The lorry is experiencing resistance forces of 900 N. The engine is driving forward with a force of 2100 N. Find the acceleration of the lorry.

Question 10. A pebble of mass 50 g is falling to solid ground through a still body of water. The pebble is experiencing a downwards acceleration of 2.5 ms^{-2}. Find the total resistive forces acting on the pebble.

Question 11. A car of mass 2000 kg reduces its speed from 18 ms^{-1} to rest in 5 seconds, along a straight flat road.

a) Find the breaking force required to achieve this.

b) State the modelling assumption you have made in calculating the breaking force.

Question 12. A vehicle of mass 900 kg reduces its speed from 20 ms^{-1} to rest in 6 seconds, along a straight flat road. It experiences a resistive force of 450 N during this time. Find the breaking force of the vehicle.

Question 13. Prerequisite knowledge for Q13-Q18: Constant Acceleration formulae A stone of mass 400 g is dropped from a bridge and falls vertically. It hits the ground 1.5 seconds after being released. A constant resistance to motion of the stone is given to be 2 N. Find the height from which the stone is dropped.

Question 14. A body of weight 20 N is accelerating along rough horizontal ground, being pushed by a force of 7 N acting parallel to the ground. A constant resistance force of 5.5 N acts against the particle.

a) Find the mass of the particle.

b) Given the particle starts at rest, find its speed after 10 seconds.

Question 15. A block of mass 3 kg starts at rest and is accelerating across a smooth horizontal ground by a force of 1.5 N acting parallel to the ground. Find how far it has travelled in the first 4 seconds of motion.

Question 16. A car of mass 1650 kg is driving with constant speed 10 ms^{-1} when it begins accelerating. After 30 m it reaches a speed of 18 ms^{-1}. Find the resultant force acting on the car during the period of acceleration.

Question 17. A car of mass 2000 kg is driving at 47 km h^{-1} when the driver notices an accident ahead. When the car is 20 metres away from the accident, the driver applies the brakes creating a total resistance force of 10 000 N. Determine whether the car stops before the accident.

Question 18. A van of mass 2250 kg drives a distance of 40 metres along a straight road, accelerating uniformly from rest to 25 ms^{-1}. Find

a) the acceleration of the van.
b) the magnitude of the force exerted by the engine, stating any modelling assumptions you have made in this calculation.
c) the magnitude of the normal reaction between the van and the road.